REINFORCE

Monte-Carlo Policy Gradient and Variance Reduction

Reinforcement Learning

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1 Setup

We work in the episodic setting. The objective and its gradient are quoted from the policy gradient theorem reports:

$$J(\theta) = \mathbb{E}_{\tau\sim p_\theta}\big[\, G_0 \,\big], \qquad \nabla_\theta J(\theta) = \mathbb{E}_{\tau\sim p_\theta}\Big[\, \sum_{t=0}^{T-1} \gamma^t \, G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t) \,\Big].$$

We want the maximizer $\theta^\star = \arg\max_\theta J(\theta)$. If the exact gradient were available, we would climb to it by gradient ascent,

$$\theta \leftarrow \theta + \alpha \, \nabla_\theta J(\theta).$$

It is not available in closed form, so this report replaces $\nabla_\theta J(\theta)$ by a sampled estimator $\widehat{g}$ and reads each estimator through the decomposition $\widehat{g}_k = g_k + b_k + \varepsilon_k$. We develop two Monte-Carlo estimators, both unbiased ($b_k = 0$): plain REINFORCE (high variance), then the same estimator with a baseline (a control variate that cuts the variance for free).

Part I REINFORCE (Reward-to-Go)

2 The Estimator

2.1 Why We Cannot Take the Exact Step

From the setup, the ideal update is exact gradient ascent $\theta \leftarrow \theta + \alpha \, \nabla_\theta J(\theta)$. Writing the gradient out,

$$\theta \leftarrow \theta + \alpha \, \mathbb{E}_{\tau\sim p_\theta}\Big[\, \sum_{t=0}^{T-1} \gamma^t \, G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t) \,\Big],$$

and the obstruction is visible: the step direction is an expectation over every trajectory the policy could generate, which we cannot compute. We do not know the dynamics, and even if we did the sum over trajectories is intractable. So we cannot take this step as written.

2.2 Replacing the Expectation by One Sample

We do have one thing: we can *run* the policy and watch what happens. Running π_θ from a start state produces a single trajectory,

$$\tau = (S_0, A_0, R_1, S_1, \ldots, S_{T-1}, A_{T-1}, R_T, S_T) \;\sim\; p_\theta,$$

which is one draw from the very distribution the expectation averages over. The Monte-Carlo move is to keep the integrand on that single draw and drop the expectation, giving the estimator

Key Result

$$\widehat{g} = \sum_{t=0}^{T-1} \gamma^t \, G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t), \qquad G_t = \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}.$$

Every piece of $\widehat{g}$ is now something the trajectory hands us: the returns G_t are the discounted sums of the rewards we observed, and the scores $\nabla_\theta \log \pi_\theta(A_t \mid S_t)$ are computed from the policy at the states and actions we actually visited. The weight G_t is a sampled return, available only once the episode ends, which is what makes REINFORCE a Monte-Carlo method.

2.3 The Stochastic Ascent Step

By construction $\mathbb{E}[\widehat{g}] = \nabla_\theta J(\theta)$: on average $\widehat{g}$ points along the true gradient. So we take the ascent step with $\widehat{g}$ in place of the gradient we cannot compute,

$$\theta \leftarrow \theta + \alpha\,\widehat{g},$$

which is *stochastic* gradient ascent: a step along a noisy sample whose mean is the exact gradient. One pass of the method is therefore: run the policy to get a trajectory, read off the returns and scores to form $\widehat{g}$, take one step, repeat.

Algorithm 1: REINFORCE (per-trajectory)

Input: policy π_θ, step size α, discount γ
initialize θ
repeat
  generate $\tau = (S_0, A_0, R_1, \ldots, A_{T-1}, R_T, S_T) \sim \pi_\theta$
  for $t = 0, \ldots, T-1$ **do**
    $G_t \leftarrow \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}$
  $\widehat{g} \leftarrow \sum_{t=0}^{T-1} \gamma^t G_t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$
  $\theta \leftarrow \theta + \alpha\,\widehat{g}$
until *converged*

2.4 Per-Timestep Form

Algorithm 1 waits until the episode ends, forms the whole sum $\widehat{g}$, and takes one step. But $\widehat{g}$ is already a sum over timesteps, so we can apply each term as it is computed. Writing $\widehat{g}_t := \gamma^t G_t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$, the two orders give the same result: at fixed θ, stepping with each $\widehat{g}_t$ in turn accumulates

$$\theta_T = \theta_0 + \alpha \sum_{t=0}^{T-1} \widehat{g}_t = \theta_0 + \alpha\,\widehat{g},$$

the single per-trajectory step. They differ only in *when* θ moves: applying updates within the episode makes step t use θ_t instead of θ_0, an $O(\alpha^2)$ discrepancy. This is the form Sutton states.

Algorithm 2: REINFORCE (per-timestep, Sutton)

Input: policy π_θ, step size α, discount γ
initialize θ
repeat
  generate $\tau = (S_0, A_0, R_1, \ldots, A_{T-1}, R_T, S_T) \sim \pi_\theta$
  for $t = 0, \ldots, T-1$ **do**
    $G_t \leftarrow \sum_{k=0}^{T-1-t} \gamma^k R_{t+k+1}$
    $\theta \leftarrow \theta + \alpha\, \gamma^t G_t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$
until *converged*

The γ^t carries two distinct discounts. Inside $G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$, the discount weights future rewards relative to step t. *The γ^t* multiplying *the term weights the visit to S_t itself: reaching S_t at time t counts with weight γ^t, the discounted state-visitation weight from the policy gradient theorem.*

3 Properties

The estimator is unbiased, and its only defect is variance. Both facts set up Part II.

3.1 Bias

Taking the expectation over the trajectory recovers the exact gradient, since $\widehat{g}$ is the integrand of $\nabla_\theta J(\theta)$ evaluated on one draw:

$$\mathbb{E}_{\tau\sim p_\theta}[\widehat{g}] = \mathbb{E}_{\tau\sim p_\theta}\Big[\sum_{t=0}^{T-1} \gamma^t G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t)\Big] = \nabla_\theta J(\theta).$$

So $b_k = 0$: the noise $\varepsilon_k = \widehat{g} - \nabla_\theta J$ is zero-mean and there is no bias term.

3.2 Variance

Since $\widehat{g}$ is unbiased, its variance is what governs the quality of the estimate, and it is the one quantity Part II attacks. Each term $\gamma^t G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$ is a scalar return times a score vector, both random through the trajectory, and the return is the noisy factor: G_t sums the entire random future, so a single draw sits far from its mean $V^{\pi_\theta}(S_t)$, and the term inherits that spread. Summed over the episode, the variance grows with the horizon and the reward scale.

The waste is concrete. G_t is large and fluctuating even when the action at step t was unremarkable, because it carries the whole downstream outcome, most of which had nothing to do with that action. Part II removes exactly this: it replaces G_t by $G_t - V^{\pi_\theta}(S_t)$, how much better than average the outcome was, which fluctuates far less while leaving the mean untouched.

3.3 Stochastic-Approximation Reading

The update $\theta_{k+1} = \theta_k + \alpha_k \widehat{g}_k$ is a Robbins-Monro recursion: unbiased drift $\nabla_\theta J(\theta_k)$, zero-mean noise ε_k, fixed point at a stationary point $\nabla_\theta J(\theta) = 0$. Under the standing step-size conditions it converges there; the high variance is what governs the rate.

Part II REINFORCE with a Baseline

4 Reducing the Variance

Part I ended with one defect: the per-timestep term $\widehat{g}_t = \gamma^t G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t)$ is unbiased, but the raw return G_t makes it high-variance. The game of Part II is to cut that variance without disturbing the mean, so the estimator stays unbiased. The tool is a *control variate*, and the specific one we use is a *baseline*.

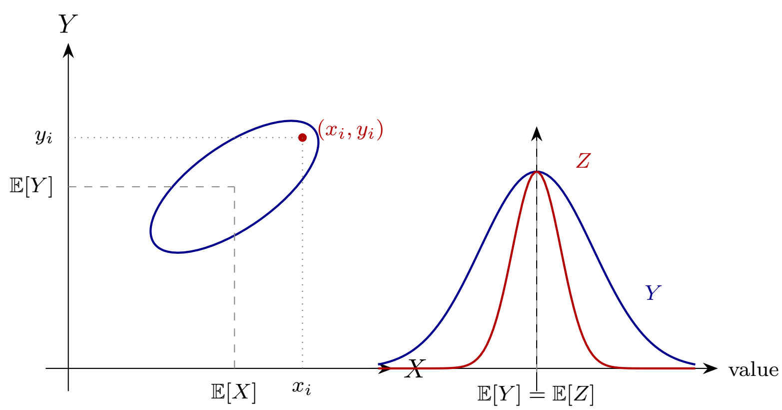
5 What a Control Variate Is

A control variate reduces the variance of an estimate by subtracting off a piece we can predict. Suppose we want $\mathbb{E}[Y]$ but Y is noisy. If we have a second quantity X that is correlated with Y and whose mean $\mathbb{E}[X]$ we *know*, we can subtract the part of Y that X predicts: what we remove has known mean, so the average is untouched, but the fluctuation shrinks. We trade noise we can explain for a quantity we already know. Full derivation in the Monte-Carlo Statistical Methods report.[1]

The mechanism in one example. We want the average *height* of a population and already know its average *weight*. On each sampled person we record both, $(x_i, y_i) = (\text{weight}, \text{height})$. Height and weight are correlated, so if a sampled person's weight lands above the known average weight, their height is probably above average too. We use that: subtract the weight-predicted part of the height deviation. The average is untouched (we subtract only a known-mean quantity), but the spread shrinks. Stronger correlation, more removed.

5.1 A Picture

Left: a one-sigma contour of $(X, Y) = (\text{weight}, \text{height})$, the tilt is the correlation. Right: the corrected variable Z sits at the same mean as Y but far narrower.



5.2 The Construction

Decompose Y into its best linear predictor from X plus an uncorrelated residual, $Y = aX + b + \epsilon$, $a = \mathrm{Cov}(X, Y)/\mathrm{Var}(X)$. Centering and subtracting the predictable part gives the corrected variable:

Key Result

$$Z = Y - a\,(X - \mathbb{E}[X]), \qquad \mathbb{E}[Z] = \mathbb{E}[Y], \qquad \mathrm{Var}(Z) = (1-\rho^2)\ \mathrm{Var}(Y), \qquad \rho = \mathrm{Corr}(X, Y).$$

The mean is unchanged, since the subtracted term has known mean zero. The variance is scaled by $(1 - \rho^2)$: the correlation ρ is the fraction of spread we can explain and remove. The three requirements: X correlated with Y, with known mean $\mathbb{E}[X]$, sampled jointly with Y.

[1] `https://github.com/SaiSampathKedari/MonteCarlo-Statistical-Methods/blob/master/reports/ch04_01_ControlVariate_Foundations-and-Intution.pdf`

6 The Baseline: One Control Variate per Timestep

We apply this construction to REINFORCE. The estimator is a sum of per-timestep terms, so we treat each term as the noisy quantity Y and give it its own control variate X. For a state-only function b, at each step t set

$$Y_t = \gamma^t G_t \, \nabla_\theta \log \pi_\theta(A_t \mid S_t), \qquad X_t = \gamma^t \, b(S_t) \, \nabla_\theta \log \pi_\theta(A_t \mid S_t).$$

X_t is sampled on the same trajectory as Y_t and shares the score, so it is correlated with Y_t. Its mean we work out next.

6.1 The Control Variate Has Known Mean Zero

The mean of X_t is exactly zero, conditionally on the state, so unlike the general case we do not even need to estimate $\mathbb{E}[X_t]$:

$$\mathbb{E}_{A_t|S_t}\big[X_t\big] = \gamma^t \, b(S_t) \, \mathbb{E}_{A_t|S_t}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\big] = 0,$$

because the expected score is zero, $\mathbb{E}_{A_t|S_t}[\nabla_\theta \log \pi_\theta(A_t \mid S_t)] = 0$. Two things make this work: $b(S_t)$ depends on the *state only*, so it pulls out of the expectation over the action; and the expected score vanishes.

> ***Why the expected score is zero (mass conservation).*** *The policy is a probability distribution over actions, so its total mass is* 1 *for every* θ. *Moving* θ *by* $d\theta$ *shifts probability between actions but the total stays pinned at* 1. *The expected score asks, on average, how much mass a nudge* $d\theta$ *creates; since the total can never move off* 1, *that average is zero.*

6.2 The Corrected Estimator

With $\mathbb{E}[X_t] = 0$ known, the corrected variable $Z = Y - a(X - \mathbb{E}[X])$ specializes to

$$Z_t = Y_t - a\,\big(X_t - \mathbb{E}[X_t]\big) = Y_t - a\,X_t = \gamma^t \big(G_t - a\,b(S_t)\big) \nabla_\theta \log \pi_\theta(A_t \mid S_t).$$

It is unbiased for the same reason as before, $\mathbb{E}[Z_t] = \mathbb{E}[Y_t]$, since the subtracted term has mean zero. The coefficient a and the function b are both ours to choose, so we fold a into b and write the corrected per-timestep term as

Key Result

$$Z_t = \gamma^t \big(G_t - b(S_t)\big) \nabla_\theta \log \pi_\theta(A_t \mid S_t).$$

The variance drop is governed by how well $b(S_t)$ predicts the return G_t: the higher the correlation, the more of G_t's spread is removed. The remaining question is which b to use.

6.3 Which Baseline: the Value Function

Among all valid state-only baselines we take

$$b(S_t) = V^{\pi_\theta}(S_t),$$

the value function. The geometric reason: picture the states on a plane with V^{π_θ} a surface over them. Fix a state s. Many trajectories pass through s, each collecting a *different* random return G_t from s onward, so above s the returns form a vertical cloud, one dot per trajectory. By definition $V^{\pi_\theta}(s) = \mathbb{E}[G_t \mid s]$ is the *center* of that cloud: the surface threads through the middle of the scatter at every state.

Subtracting the surface recenters the cloud to zero. Without it, a high-value state lifts the whole cloud, so every trajectory through s carries a large weight of the same sign, an offset that is there because the *state* was good, not the *action*, and is pure noise for judging the action. After subtracting, the weight $G_t - V^{\pi_\theta}(S_t)$ measures each trajectory's dot from its cloud center: trajectories that beat the local average get a positive weight, those that fell short a negative one. The mean is untouched, but the weight shrinks from "around $V^{\pi_\theta}(s)$" to "around zero," and since variance scales with the size of the weight, it drops. The centered weight $G_t - V^{\pi_\theta}(S_t)$ is an unbiased sample of the *advantage* $A^{\pi_\theta}(S_t, A_t) = Q^{\pi_\theta}(S_t, A_t) - V^{\pi_\theta}(S_t)$: how much better this action did than the state's average.

The numbers. *Suppose* $V^{\pi_\theta}(s) = 100$ *and trajectories through* s *return* $95, 103, 98, 106, \ldots$ *(spread of a few units around* 100*). The raw weights* G_t *are all near* 100*: large, same sign, every trajectory pushes the action up by roughly the same big amount. Centered, the weights* $G_t - V^{\pi_\theta}(s)$ *become* $-5, +3, -2, +6$*: small, both signs, each saying how much this action beat or missed the local average. Same information about the action, but the weight has gone from "around* 100*" to "around* 0*," and the variance with it.*

7 Where This Goes Next

Two pieces are still open, and both are the entry point to the next algorithm rather than to this one.

First, we do not know V^{π_θ}, so it must be learned. Fitting a value predictor $\widehat{V}_w(s)$ by regression on the observed returns (S_t, G_t) turns the baseline into a second estimator running alongside the policy. An imperfect fit stays safe: the mean-zero identity holds for *any* state-only function, so the gradient remains unbiased regardless of the fit, and a poor fit merely removes less variance.

Second, once a learned value function is on hand, it can supply more than a baseline: it can replace the sampled return G_t in the weight itself, trading variance for a little bias. That is precisely the move that defines *actor-critic*, where the learned value function becomes the critic. We carry the control-variate trick, and the advantage $G_t - V^{\pi_\theta}(S_t)$ it produces, into that report.